

PROJECT SUDHARSHAN

Autonomous Multi-Vector Dark Web Threat Actor Attribution & Intelligence Platform

The Complete Master Reference & Engineering Blueprint: Plain-English Conceptualization, Algorithmic Deep Dives, Full Forms & Glossary, 6-Stage Intelligence Pipeline, Database Models, and Court-Admissible Electronic Evidence Standards.

Team:	CodeMaverick 0001
Theme:	Blockchain & Cybersecurity
Target Agency:	National Technical Research Organisation (NTRO) / LEAs
System Stack:	Next.js 16, Python FastAPI, PostgreSQL 16, Neo4j 5, Tor Daemon
Release Version:	1.0.0-MVP (Fully Offline / Air-Gap Tested)

JURY & EVALUATION PHILOSOPHY

Zero Unverified Claims: This project strictly rejects speculative "90% AI magic" numbers. Sudarshan establishes attribution purely through deterministic, reproducible mathematical heuristics, verifiable cryptographic proof, and human-in-the-loop judicial standards (Section 65B Indian Evidence Act).

1. Foundational Glossary & Full Forms (A to Z)

To understand the problem and solution deeply, every core cybersecurity concept and acronym must be precisely understood:

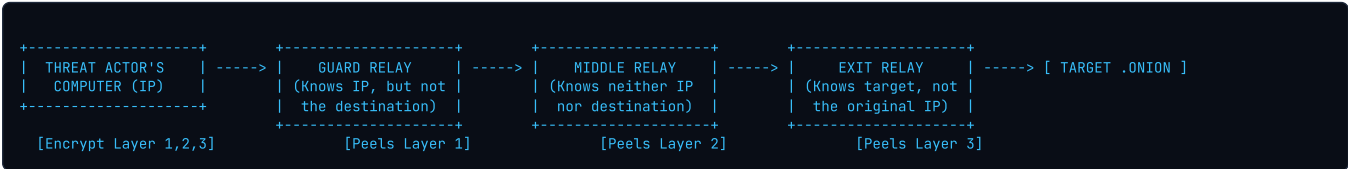
Term / Acronym	Full Form	Plain-English & Technical Explanation
Tor	The Onion Router	An open-source anonymity network that directs internet traffic through a worldwide, volunteer overlay network consisting of thousands of relays. Traffic is encapsulated in 3 nested layers of encryption (like an onion), ensuring no single relay knows both the original source and destination.
Threat Actor	<i>Industry Standard Term</i>	An individual or syndicate that conducts malicious cyber activities (e.g. ransomware extortion, data theft, weaponized exploit sales, state-sponsored cyber espionage). In dark web contexts, they hide behind handles like Shadow99 or DarkVendor .
Attribution	<i>Intelligence Methodology</i>	The scientific process of tracking a digital persona, alias, or cyberattack back to its real-world human perpetrator, physical location, or sponsoring nation-state organization.
De-Anonymization	<i>Forensic Process</i>	Unmasking the true technical identity (clearnet IPv4/IPv6 address, real email, physical ISP subscriber) of an anonymous network participant.
PGP	Pretty Good Privacy	An encryption program created by Phil Zimmermann in 1991 that provides cryptographic privacy and data authentication. It uses Asymmetric Cryptography (Public Key for encrypting/verifying, Private Key for decrypting/signing).
PGP Fingerprint	<i>160-bit Hex Digest</i>	A unique 40-character hexadecimal string (e.g. 9A4F 88C1 204B ... CD49) that uniquely identifies a specific PGP public key. In dark web forums, hackers sign pastes with this key to prove their reputation.
Cleartnet	<i>Clear / Surface Network</i>	The standard, non-encrypted public internet (Google, GitHub, Reddit) indexed by traditional search engines and routed via regular Domain Name System (DNS) and public IP addresses.
.onion Hidden Service	Tor Onion Service (v3)	A web server configured to be reachable exclusively within the Tor network. It uses a 56-character base32 address derived directly from its public key (e.g. exp ... 56chars.onion). It has no public DNS entry and its hosting IP is concealed.
SOCKS5	Socket Secure (Version 5)	An internet protocol that exchanges network packets between a client and server via a proxy server. Sudarshan uses SOCKS5 (socks5h : // 127 . 0 . 0 . 1 : 9050) to route crawler requests into the Tor network.
IOC	Indicator of Compromise	Forensic artifacts that identify rogue activity: crypto wallet addresses, PGP keys, malicious IPs, hashes, email aliases, or leaked SSL certificates.
SHA-256	Secure Hash Algorithm (256-bit)	A cryptographic hash function that outputs an unforgeable 64-character hexadecimal digest. If even a single byte of a web page is changed, the SHA-256 digest changes completely. Used to guarantee evidence integrity.
Stylometry	<i>Linguistic Authorship Analysis</i>	The statistical study of written text style (average sentence length, punctuation frequency, vocabulary richness, characteristic spelling errors) to identify whether two different handles were written by the same human author.
Diurnal Activity	<i>Circadian Temporal Profiling</i>	Analyzing the 24-hour UTC distribution of an actor's timestamps to identify sleep and active windows, allowing investigators to calculate their most probable geographical timezone (e.g. UTC+05:30 for India).
Section 65B	Section 65B, Indian Evidence Act (Sec 63 BSA)	The mandatory statutory rule in Indian law governing the admissibility of electronic evidence. Requires certification of system integrity, NIST timestamps, and immutable cryptographic hashes to prevent evidence tampering.
UTXO / Co-Spend	Unspent Transaction Output	Bitcoin's accounting model. The <i>Co-Spending Heuristic</i> states that when a transaction combines multiple input addresses to fund an output, all those input addresses are controlled by the same actor.

2. The Core Problem: Why Dark Web Attribution is So Difficult

THE ONION ROUTING DILEMMA

On the surface web (Clearnet), when you visit a website, your computer connects directly to the server's IP address. Law enforcement can simply subpoena the internet service provider (ISP) to find who was assigned that IP address at that exact second.

The **Tor network destroys this connection** using 3-hop layered encryption:



- **Guard Relay:** Sees the user's real IP address, but has no idea what website the user is trying to reach.
- **Middle Relay:** Only sees the Guard and Exit nodes; completely blind to both the user and the final destination.
- **Exit Relay:** Sees the target website, but only knows the Middle Relay's IP address, never the true originator.

Furthermore, **Tor Hidden Services (.onion)** don't even use Exit Nodes—both the user and the server rendezvous at an encrypted midpoint inside the network. Traditional network sniffing, IP geolocation, and packet analysis are rendered mathematically impossible.

The Breakthrough: Human Error & Operational Security (OpSec) Failures

Tor's cryptography is robust, but the humans who operate inside it are fallible. Threat actors conduct business over months and years, leaving behind trails of digital breadcrumbs. Attribution is achieved not by cracking Tor's math, but by **correlating multiple independent operational errors** across distinct intelligence vectors.

3. How Sudarshan Works: The "Easy Way" vs. "Hard Way"

Sudarshan deploys **four deterministic de-anonymization vectors** coupled with an explainable scoring algorithm:

VECTOR 1: PGP Key Fingerprint Reuse

The Easy Way: Imagine a criminal who uses a fake name in every city, but always stamps his letters with the exact same personalized wax seal. Once police find that same wax seal on a university application from 5 years ago, his fake names collapse.

The Hard Way: Scraper extracts `-----BEGIN PGP PUBLIC KEY BLOCK-----` blocks, parses them using `PGPy`, calculates the 160-bit SHA-1 Key ID / fingerprint, and cross-references public keyservers (`keys.openpgp.org` , SKS pools, GitHub GPG commits). A hit reveals verified clearnet email addresses and real developer handles.

VECTOR 2: Crypto Flow & Exchange Clustering

The Easy Way: Stolen cash inside a hidden safe is useless unless the criminal spends it. Eventually, the criminal sends Bitcoin to buy a car or deposit into a bank account linked to his government ID.

The Hard Way: Regex extracts BTC (Legacy, SegWit, Bech32), ETH (0x hex), and XMR addresses. Multi-input UTXO co-spending heuristics cluster multiple addresses under one actor. Graph traversal follows transactions across hops until they hit indexed deposit addresses belonging to KYC-compliant exchanges (Binance, CoinDCX, WazirX) where legal warrants can be issued.

VECTOR 3: Infrastructure Leaks & Misconfigurations

The Easy Way: A thief builds a secret hideout in the jungle, but forgets to take down the official electricity meter box on the front door that displays his home address.

The Hard Way: Tor hidden service operators make Nginx/Apache configuration blunders: (1) Reusing SSL/TLS certificates on both clearnet and `.onion` hosts; (2) Leaking real IP in response headers or error tracebacks; (3) Loading hardcoded clearnet assets. Our scanner extracts cert serials, calculates Favicon MurmurHash3, and matches them against Censys/Shodan datasets to reveal the origin IP.

VECTOR 4: Stylometry & Diurnal Timeline

The Easy Way: Even if a criminal gets plastic surgery and changes his name, his handwriting habits, common typos, and the hours he sleeps remain identical.

The Hard Way: For rebranded personas who changed both wallet and PGP key, Sudarshan computes dense sentence embeddings using Sentence-Transformers (`all-MiniLM-L6-v2`), calculates lexical richness (Yule's K), punctuation distribution, and plots a 24-hour UTC activity histogram to establish authorial match and geographic timezone.

The Mathematical Confidence Scoring Engine (Explainable, No Black Box)

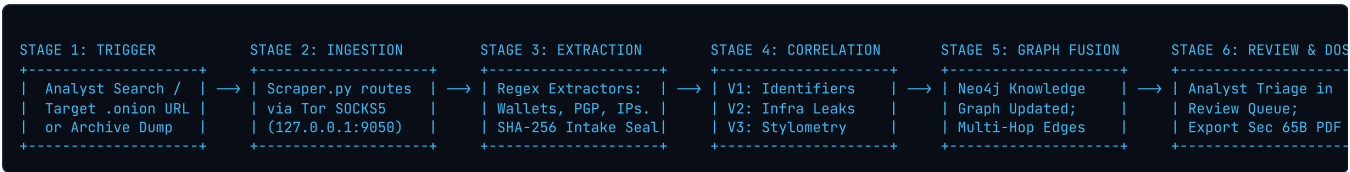
Unlike black-box AI systems that output unverifiable percentages, Sudarshan calculates confidence using a transparent, additive points model:

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Confidence Score = Σ (Vector_Weight_i × Evidence_Strength_i)

- Vector 1: Shared PGP Fingerprint Match      = +35 Points
- Vector 2: Clearnet IP / SSL Serial Leak      = +30 Points
- Vector 3: Crypto Co-Spend / KYC Link        = +25 Points
- Vector 4: Stylometry & Diurnal Alignment    = +10 Points

Tiers: High Confidence (≥ 75%) | Medium Confidence (40% - 74%) | Lead Only (< 40%)
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4. The End-to-End Intelligence Pipeline (6 Stages)



- 1. **Stage 1 (Target Trigger):** The analyst enters a target .onion URL or uploads a sanitized dark web forum dump archive.
- 2. **Stage 2 (Tor-Routed Ingestion):** Scraper.py establishes connection through Tor SOCKS5 proxy (127.0.0.1:9050) with circuit isolation and anti-fingerprint header rotation.
- 3. **Stage 3 (Extraction & Evidence Integrity):** Raw HTML text is immediately stamped with a SHA-256 cryptographic digest. Regex extractors parse BTC, ETH, XMR wallets, PGP key blocks, emails, and exposed IP strings.
- 4. **Stage 4 (Multi-Vector Correlation):** The correlation engine checks for shared PGP fingerprints, wallet clusters, and infrastructure overlaps against the historical database.
- 5. **Stage 5 (Graph Fusion):** Results are transformed into graph adjacency nodes and edges within Neo4j and relational records in PostgreSQL.
- 6. **Stage 6 (Human Review & Legal Dossier):** The analyst reviews the evidence trail in the Review Queue (/links), confirms the attribution, and generates a Section 65B certified PDF dossier ready for judicial submission.

5. System Architecture & Technical Stack

Layer	Technology Selected	Version	Role & Responsibility
Frontend UI	Next.js (React 19)	16.3.3	Fast, reactive, responsive cyber defense console; client-side graph interaction.
Styling	Tailwind CSS	4.3.3	High-contrast HackBIOS dark theme tokens, responsive grids, zero styling bloat.
Graph Canvas	react-force-graph-2d	1.29.2	HTML5 canvas force physics; smooth rendering of 500+ nodes and multi-hop edges.
Motion Engine	Motion.dev (motion)	14.0.0	Physics-based drawer transitions, live pulse badges, and card animations.
Backend API	Python FastAPI	0.110+	Asynchronous high-throughput REST API with OpenAPI Swagger documentation at /docs .
Relational DB	PostgreSQL (SQLite dev)	16-alpine	Transactional source of truth; immutable audit logs; full-text document search.
Graph Database	Neo4j Community	5.18+	Graph topology engine; fast Cypher queries for multi-hop link traversal.
Task Scheduler	Celery + Redis	5.3 / 7.0	Background job queue for Tor crawling, periodic scans, and batch correlation.
Tor Client	Tor Daemon + stem	0.4.8+	SOCKS5 proxy routing (127.0.0.1:9050) and circuit renewal control.

The Dual-Database Advantage: Why PostgreSQL + Neo4j?

- **PostgreSQL:** Relational tables ensure ACID compliance, strict foreign-key referential integrity, and append-only audit logs that cannot be modified by defense lawyers in court.
- **Neo4j:** In relational SQL, traversing 3 or 4 hops (Actor A → Wallet 1 → Transaction → Wallet 2 → Actor B) requires expensive recursive table joins. In Neo4j, graph pointers traverse 4 hops in **sub-millisecond latency**.

6. The 10 Analyst Webpages (UI Information Architecture)

To eliminate investigator fatigue and sensory overload, Sudarshan divides intelligence operations across 10 clean, purposeful screens:

Route	Page Name	Key Features & Actions
/	Executive Overview	4 KPI cards (Monitored Actors, Linked Entities, IP Leaks, Network Status); Priority Surveillance Grid; Ingestion Provenance badge.
/graph	Interactive Graph Canvas	Full-screen 2D force graph; custom node coloring (Actor, Wallet, IP, PGP); right-hand slide-over Inspector Drawer; node search.
/actors	Threat Actor Directory	Searchable table with handle, category, primary crypto wallet, PGP fingerprint, confidence score, and one-click jump to graph view.
/actors/:id	Actor Deep-Dive Dossier	Exhaustive entity profile: identified wallets, cryptographic public keys, raw forum post evidence excerpts, and investigator case notes.
/links	Persona Review Queue	Human-in-the-loop review deck showing candidate persona pairs (Persona A ↔ Persona B), score breakdown, and [Confirm Link] / [Reject Lead] actions.
/infrastructure	Infrastructure Scanner	Vector 2 table displaying Tor hidden services leaking clearnet host IPs via SSL cert serials, banners, and Favicon hashes.
/timeline	Timeline & Historical Search	Date-range filter, full-text keyword query, and interactive 24-hour UTC diurnal posting histogram for timezone estimation.
/reports	Legal Dossier Generator	Section 65B Indian Evidence Act compliant report with SHA-256 evidence integrity seals, statutory header, and one-click PDF export.
/system	System Ops & Audit Trail	Crawler launch console (Scraper.py), live Tor SOCKS5 status monitor, and immutable officer audit log table.
/login	Officer Authentication	Secure gateway with Officer Badge ID, master password, and offline 6-digit TOTP authenticator code input.

7. Frequently Asked Questions (FAQ) — For Teammates & SIH Judges

Q1: CAN TOR ENCRYPTION ACTUALLY BE BROKEN BY SUDARSHAN?

Answer: No, and no credible cybersecurity tool claims to break Tor's underlying RSA/AES onion routing cryptography. Sudarshan exploits *operational security (OpSec) errors* made by human operators: PGP key reuse across clearnet/darknet, cryptocurrency co-spending into KYC-compliant exchanges, server misconfigurations leaking origin IPs, and stylometric linguistic patterns.

Q2: WHAT IF THE CRIMINAL USES A CRYPTOCURRENCY TUMBLER OR MIXER (E.G. TORNADO CASH, CHIPMIXER)?

Answer: Mixers break 1-to-1 direct transfer links, but they do not eliminate forensic visibility: (1) Many actors bypass mixers due to 2-5% commission fees or mixer exit delays; (2) Time-volume statistical correlation can link mixer inputs with mixer outputs; (3) When the actor eventually cashes out to fiat currency, they must interface with a KYC exchange where law enforcement subpoenas can be served.

Q3: HOW RELIABLE IS STYLOMETRY? CAN A THREAT ACTOR FAKE THEIR WRITING STYLE?

Answer: Stylometry is treated strictly as a *supporting signal* (+10 points max), never as standalone proof. While an actor can intentionally alter words or use translation tools in a few posts, cognitive fatigue makes it nearly impossible to sustain unnatural sentence cadence, punctuation frequencies, and syntactic rhythm over months of active posting.

Q4: WHY DOES THE SYSTEM ENFORCE A "HUMAN IN THE LOOP" REVIEW QUEUE?

Answer: Autonomous AI systems that output automated accusations create unacceptable legal liability. Under Indian criminal procedure, a court cannot admit a machine-generated accusation without human verification. Sudarshan acts as an intelligent decision-support system: the engine proposes candidate links, but an authorized intelligence officer must review the corroborating evidence and certify the link.

Q5: WHAT MAKES SUDARSHAN'S DOSSIER LEGALLY ADMISSIBLE UNDER SECTION 65B OF THE INDIAN EVIDENCE ACT?

Answer: Section 65B (and Section 63 of Bharatiya Sakshya Adhiniyam, 2023) requires: (1) Proof of uninterrupted system operation; (2) Prevention of electronic record tampering; (3) Cryptographic hash integrity. Sudarshan computes a SHA-256 hash at the exact second a web page is scraped, logs NIST timestamps and Tor circuit metadata, and binds them into an immutable audit trail stamped on the final PDF dossier.

Q6: ARE WE CRAWLING LIVE ILLEGAL DARK WEB MARKETPLACES DURING THE HACKATHON DEMONSTRATION?

Answer: No. Live crawling of illicit markets during public demonstrations violates legal ethics and venue safety rules. Our demonstration runs on a verified 10-actor synthetic ground-truth corpus (M18) and sanitized academic research archives (e.g. Gwern darknet market dumps). The live Tor SOCKS5 crawler module exists and is verified on safe test onion services for deployment in authorized law enforcement environments.

END OF TECHNICAL SPECIFICATION — PROJECT SUDHARSHAN // NTRO SIH 2026